

Xu & Xie (2023)

Conformal Prediction for Time Series

(EnbPI, IEEE TPAMI)

Paper Overview

1. Why this paper exists

Standard conformal prediction (CP) for regression requires either (a) holding out a calibration set (split conformal) or (b) refitting the predictor once per candidate y (full conformal). Neither plays well with time series. Splitting a time series throws away the most recent data; refitting an arbitrary forecasting model (LSTM, GBM) per candidate is computationally infeasible. The plain jackknife is asymptotic-only and requires a stability assumption that's rarely satisfiable for modern ML predictors.

Key Insight

The bootstrap-LOO trick. Xu and Xie (2023a) exploit a structural observation: a bootstrap ensemble *already contains leave-one-out information*. For each training index i , the bootstrap models whose training sets did not include index i jointly form a leave-one-out estimator $\hat{f}^{-i}$ — with zero extra computational cost beyond the original bootstrap fit. EnbPI rides this insight to give time-series CP that requires *no data split* and *no retraining at test time*.

The result is EnbPI (*Ensemble batch Prediction Intervals*): a wrapper around any ensemble-trainable predictor that gives distribution-free prediction intervals with asymptotic coverage guarantees under β -mixing weak dependence. The paper is published in IEEE TPAMI and is the canonical reference for what Stocker et al. (2025) call the “refresh residual pool” family of CP-for-time-series methods.

2. The bootstrap-LOO insight

The core insight is structural rather than algorithmic. A standard bootstrap ensemble trains B predictors on B resampled training sets. By the bootstrap design, each individual training point i is excluded from approximately $B/e \approx 0.368B$ of the resamples. The predictors trained on those $\approx 0.368B$ resamples constitute a *leave-one-out estimator* of $\hat{f}$ at index i :

$$\underbrace{\hat{f}^{-i}(x)}_{\text{LOO estimator at training index } i} = \frac{1}{|\{b : i \notin \mathcal{J}_b\}|} \sum_{b : i \notin \mathcal{J}_b} \underbrace{\hat{f}^{(b)}(x)}_{\text{bootstrap model } b}.$$

How This Works

Why this is "essentially free". A standard bootstrap ensemble is what you'd build for variance reduction anyway. The leave-one-out structure is a side product: it costs nothing beyond the bootstrap fit itself, no retraining required. The plain jackknife would need n separate model fits; the jackknife-after-bootstrap construction (J+aB) of Kim, Xu, and Barber Kim, Xu, and Barber (2020) obtains n LOO estimators from a single B -bootstrap fit. EnbPI extends J+aB to time series.

3. The algorithm

Given training data $\{(X_t, Y_t)\}_{t=1}^T$, ensemble size B , and target miscoverage α :

Step 1: bootstrap ensemble. Train B point predictors $\{\hat{f}^{(b)}\}_{b=1}^B$ on bootstrap resamples $\{\mathcal{J}_b\}_{b=1}^B$ of the training data.

Step 2: per-index LOO aggregation. For each training index t ,

$$\hat{f}^{-t}(x) = \frac{1}{|\{b : t \notin \mathcal{J}_b\}|} \sum_{b : t \notin \mathcal{J}_b} \hat{f}^{(b)}(x).$$

Step 3: LOO residuals.

$$\underbrace{\hat{\varepsilon}_t}_{\text{LOO residual at index } t} = \underbrace{Y_t - \hat{f}^{-t}(X_t)}_{\text{out-of-bag prediction error}}.$$

Step 4: test-time interval. For a new X_{T+h} , form

$$\underbrace{\hat{C}_h(X_{T+h})}_{\text{prediction interval at horizon } h} = \hat{f}(X_{T+h}) + \left[\underbrace{\hat{Q}_\beta(\{\hat{\varepsilon}\})}_{\text{lower endpoint}}, \underbrace{\hat{Q}_{1-\alpha+\beta}(\{\hat{\varepsilon}\})}_{\text{upper endpoint}} \right]$$

where $\beta \in [0, \alpha]$ is chosen by inner optimisation to minimise the interval width — yielding *asymmetric* intervals when the residual distribution is skewed.

Step 5: sliding-window update. As new observations arrive, append them to the residual window and (optionally) drop the oldest. *Crucially, $\hat{f}$ is never retrained.*

How This Works

Cost analysis. One ensemble fit upfront (the bootstrap cost — typically what you’d pay for variance reduction anyway). $O(1)$ per test point thereafter (just a quantile of a sliding window). This is much cheaper than full conformal ($|\mathcal{Y}|$ refits per test point) or the plain jackknife (n refits total). EnbPI’s compute profile is what makes it deployable in streaming forecasting systems.

4. Coverage results

THEOREM (Asymptotic coverage gap, informal). *Under β -mixing of (X_t, Y_t) and consistency of the bootstrap LOO estimator, both the marginal and conditional coverage gaps of EnbPI converge to zero as $T \rightarrow \infty$:*

$$\left| \mathbb{P}(Y_{T+h} \in \hat{C}_h(X_{T+h})) - (1 - \alpha) \right| \xrightarrow{T \rightarrow \infty} 0.$$

The paper also bounds the symmetric difference between EnbPI’s interval and the oracle (narrowest conditionally valid) interval, controlling efficiency in addition to validity.

What This Result Says

What’s unusual about this theorem. Conditional coverage — $\mathbb{P}(Y \in \hat{C}(X) \mid X = x) \rightarrow 1 - \alpha$ — is usually impossible at finite n in distribution-free CP (Lei–Wasserman 2014 impossibility). EnbPI obtains it *asymptotically* by paying two prices: (1) consistency of the bootstrap LOO estimator (a non-trivial assumption that effectively rules out distribution-free guarantees), and (2) β -mixing dependence on the data process. The price is worth paying because the conditional guarantee is what time-series practitioners actually need — intervals that are valid *at each time point*, not just on average across time.

Why β -mixing rather than i.i.d. β -mixing is a quantitative weak-dependence condition that holds for stationary ARMA, many GARCH variants, and finite-state Markov chains. It is strictly weaker than i.i.d. but quantitatively controllable, which is what the proof needs. The mixing rate appears in the convergence rate of the coverage gap.

What This Assumption Means

What β -mixing gives up. The guarantee fails (or degrades) on processes with long memory (e.g., fractionally integrated processes, ARFIMA with $d > 0$), non-stationary noise (e.g., unit-root processes), or strongly persistent regime-switching. In those settings, methods that adapt online (ACI Gibbs and Candès (2021); Zaffran et al. (2022)) are the safer choice — they trade EnbPI’s conditional coverage for long-run marginal coverage on *any* sequence.

5. Empirical findings, limitations, and connections

Benchmark suite. EnbPI is evaluated on five high-stakes time-series domains:

- Solar and wind energy generation (forecasting renewable supply).
- Multivariate anomaly detection (intrusion / fault detection).
- Greenhouse-gas concentration (environmental monitoring).
- Appliance energy consumption (smart-grid demand).
- Beijing air quality (PM2.5 forecasting under regime shifts).

Baselines beaten. Against ARIMA, ICP (inductive conformal prediction — vanilla split CP), WeightedICP, QOOB, AdaptCI (the original Gibbs–Candès ACI), and J+aB (Kim, Xu, Barber):

- EnbPI delivers shorter intervals at nominal coverage.
- Robust to missing data (the sliding window absorbs gaps gracefully).
- Handles change points via sliding-window length (shorter window $\rightarrow$ faster adaptation).

Why This Matters

Why the benchmark matters beyond academic publishability. Each of the five domains is a real deployment context where uncertainty calibration has direct consequences — grid operators rely on solar-wind forecasts for dispatch decisions, environmental regulators rely on GHG forecasts for policy, public-health authorities rely on PM2.5 forecasts for advisories. A method that delivers nominal coverage *and* short intervals across all five is robust to most real-world time-series structures, not just toy AR(1) simulations.

Limitations and follow-up work.

- Degrades as β -mixing slows toward long memory. For non-mixing processes, the consistency assumption fails and the conditional-coverage guarantee no longer holds.
- For abrupt distribution shift, pair with ACI (online α -update). The hybrid ACI + EnbPI inherits ACI’s any-sequence guarantee while keeping EnbPI’s conditional-coverage advantage in the stable regime.
- For heteroskedasticity, use the SPCI follow-up Xu and Xie (2023b), which replaces the empirical residual quantile with a learned quantile-regression score. SPCI handles scale-dependent residual distributions that EnbPI’s empirical-quantile approach cannot.

Connections. EnbPI anchors the “refresh residual pool” family in Stocker et al.’s Stocker et al. (2025) four-family taxonomy. Its direct predecessor is Kim, Xu, and Barber’s J+aB Kim, Xu, and Barber (2020), which establishes the bootstrap-LOO trick for the exchangeable setting; EnbPI extends it to non-exchangeable time series. The complementary “adapt- α ” family (ACI Gibbs and Candès (2021); Zaffran et al. (2022)) targets the same problem via online updates instead of residual refresh; the two families compose into hybrid methods that have become the working frontier of CP for forecasting. For applications to causal estimands, DR-ACI (Koukorinis 2026) combines EnbPI-style residual refresh with doubly-robust treatment-effect estimation under temporal dependence, building on the double-machine-learning framework of Chernozhukov et al. Chernozhukov et al. (2018).

References

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